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MST-II
CPE-202: Python Programming
01/2021

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Time Allowed: 60 Minutes **Max. Marks: 15**

Note: Papers sent only on the above email-ids will be considered. Kindly do not send on any other email-ids.

N.B.: Section **A** is *compulsory*. Attempt ***any two*** questions from Section-B.

SECTION-A

Q1 i) Discuss the concept of various 'objects' created to use with a data-frame with examples. 1

ii) A data-frame '**Employees_Address**' requires a new attribute '**PINCODE**'to be added to it. Write Python commands to add this new attribute to this data-frame. 1

iii) How will you add data value of 'PINCODE' for any specific record in the above '**Employees_Address**' data-frame. 1

iv) How will you count all the NULL values in above data-frame? 1

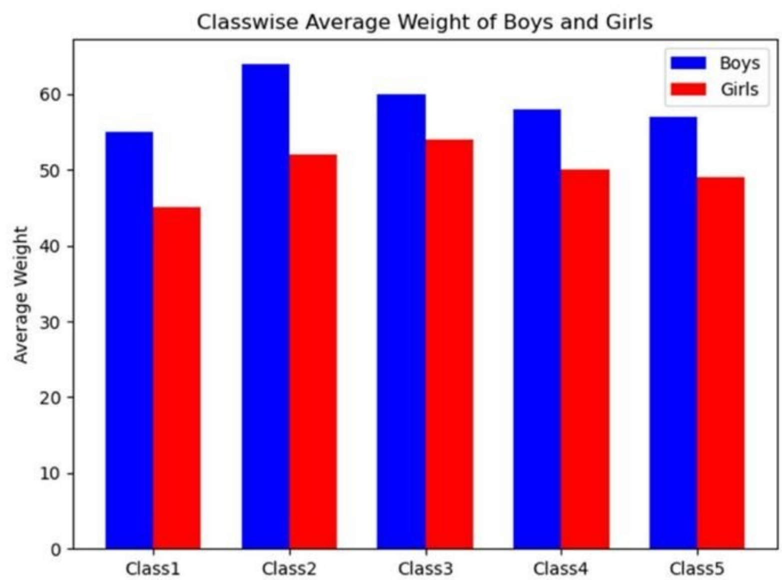
v) What is the benefit of using `inplace` parameters in data handling tasks. 1

SECTION-B

Q2. Explain how the problem of missing, duplicate and categorical values can be handled. 5

Q3. A data analytics problem required three .csv data files. How will you load all three files into a single data-frame object? Write the Python procedure for the task to justify your answer. 5

Q4. Using your own data, write Python script to plot the following figure with the given elements. 5



PYTHON PROGRAMMING
DCSE MST-II
SESSION: JULY-DECEMBER 2019

Time Allowed: 60 Minutes

MAX. MARKS: 15

SECTION-A

Question 1 is compulsory. Attempt ***any two*** from section-B.

NOTE: Your Answer-scripts should LEGIBLE-TO-READ, otherwise not to be evaluated.

Q. 1.

a) Create a Series from a list and explain its *isunique*, *ndim*, *shape* and *size* attributes 1

b) With example of each, discuss mutable and immutable structures 1

c) Give example of *sort_value*(), *sort_index*() and *astype*() with suitable data object. 1

d) What is dict()? Explain with example to use it to store data. 1

e) Give one example to plot line, pie and bar chart with data of your choice. 1

SECTION-B

Q2. Write python script to load MS-Excel file into a DataFrame and describe the use and purpose of min, max, head and tail methods with it. (5)

Q3. Give one example to delete rows and columns from a DataFrame. (5)

Q4. How the NULL values are handled in a dataset. (5)

(5)

12101018 Prem kumar

MST-II CPE-202: Python Programming
Department of Computer Science & Engineering

Time Allowed: 60 Minutes **SECTION-A (Compulsory)** **Max. Marks: 15**

Q1 i) What is broadcasting? 1

ii) Discuss the use cases when dataframe methods `iloc` and `loc` are preferred over each other. 1

iii) Write python code to insert a column named `Marks` between column `Age` and `State` in the dataframe `|Name|Age|State|Country|`. The column can have any random values. 1

iv) Write python statement to extract 'column names' from a dataframe. 1

v) What is the benefit of using `inplace` parameter in the data handling tasks. 1

SECTION-B (Attempt any two)

Q2. Explain how the problem of missing, duplicate, and categorical values can be handled. 5

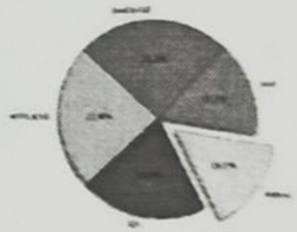
Q3. Discuss the following methods: 5

i). `idxmax` ii). `sort_values` iii). `astype` iv). `describe` v). `info`

Q4. Using the following data: 5

```
slices = [59219, 55466, 47544, 36443, 35917]
labels = ['JavaScript', 'HTML/CSS', 'SQL', 'Python', 'Java']
```

Write Python script to plot the given figure.



MST-II
CPE-202: Python Programming
Computer Sci. & Engg.
Punjabi University, Patiala

Time Allowed: 60 Minutes **Max. Marks: 15**
Date:
Note: Section **A** is *compulsory*. Attempt ***any two*** questions from Section-B.

SECTION-A

Q1 i) Explain the concept of broadcasting with the help of an example. 1
ii) Differentiate between a series and dataframe object. 1
iii) How a column can be dropped from the dataframe? Explain with the help of an example. 1
iv) Differentiate between a label and a legend? 1
v) What is the benefit of using **`inplace`** parameter in data handling tasks. 1

SECTION-B

Q2. Explain how the problem of missing, duplicate and categorical values can be handled. 5

Q3. From the following dataframe (suppose 'df'), write python code to: 5
a. Extract the columns 'Name' and 'Number'
b. Extract the 1st and 3rd rows.

Name	Team	Number	Position	Age
Avery Bradley	Boston Celtics	0	PG	25
Jae Crowder	Brooklyn Nets	99	SF	25
John Holland	New York	30	SG	27
R.J. Hunter	Toronto Raptors	28	SG	22

Q4. Using your own data, write python script to plot the following figure with the given elements. 5

CSEB2301T: Python Programming

MST-II: 11/2024

Time Allowed: 60 Minutes

N.B.: Question 1 is compulsory. Attempt any two from section-B. Marks of each question mentioned against each

Q1 5x1=5

a) Which Pandas Series methods you will use to extract data from D= {'A':10, 'B':20, 'C':30, 'D':40, 'E':50} so that it may look like data given right side. Justify your answer with examples. Also differentiate between *type* and *dtype* of above Series object.

b) Give an example to import MS-Excel file to load into Panads DataFrame and extracts any column's ... ware.

c) How will you check data cells with 'NaN' entries in a Series and DataFrame. Give example.

d) Give examples to use *loc* and *iloc* with Pandas Series and DataFrame for desired data retrieval.

e) What do you mean by broadcasting data operation. Give one example.

SECTION-B

Attempt any two questions

Q2. An 'empData.csv' file contains employee's data with EmpID, EmpFullName, EmpDesig, EmpDeptt, EmpSalary How you will create a DataFrame and increase the EmpSalary at 0.03 of all records.

Q3. Give one example to create a bar chart from the area data given in km^2
Area= {'Punjab': 50362, 'Haryana': 44212, 'Assam': 78438, 'Kerala': 38963, 'Manipur': 22327, 'HP': 55673, 'Bihar': 94163}. Also use labels, titles, legends to increase the readability of plot.

Q4 Using DataFrame given in Q.No. 2, perform the following operations:

1. Drop the column EmpDesig
2. Delete rows which are completely empty
3. Delete rows where EmpDeptt and EmpSalary are empty

DEPARTMENT OF COMPUTER SCEINEC & ENGINEEING

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MST-2 Python Programming Sem-3rd

MM: 15 Examiners: Dr. Harpreet, Dr. Navdeep and Dr. Gaurav

SECTION – A

(Each question carries 1 mark)

1. Write output of the following code:

```
import numpy as np
a = np.arange(1, 10, 2)
print(a[::-1])
```

2. Write any two attributes of a Pandas Dataframe.
3. What is the difference between `loc[]` and `iloc[]` in Pandas?
4. Write the use of "inplace" parameter.
5. What is the use of `plt.xlabel()` and `plt.ylabel()` functions?

SECTION – B

(Each question carries 5 marks) (Attempt any two)

1. Write a Python program to read data from a CSV file into a DataFrame, delete a row, and sort the DataFrame using the `sort_values()` method.
2. Explain different types of charts in Matplotlib (Line, Bar, Scatter, Pie, Box) with suitable examples.
3.

```
import pandas as pd
data = {
    'Product': ['Laptop', 'Tablet', 'Laptop', 'Mobile', None, 'Tablet', 'Laptop'],
    'Price': [70000, 25000, 70000, None, 15000, 25000, 70000],
    'Quantity': [3, 5, 3, 2, 1, None, 3]
}
df = pd.DataFrame(data)
print(df)
```

- Identify all missing values and display the total count per column.
- Remove all duplicate rows based on all columns.

MST-2

CPE-202: Python Programming

Department of Computer Science & Engineering

Time Allowed: 60 Minutes

Max. Marks: 15

Note: Section A is compulsory. Attempt *any two* questions from Section-B.

SECTION-A

Q1
iv) Write a code snippet in Pandas to sort a DataFrame based on a specific column in ascending order. 1
v) What for are idxmax and idxmin used? 1
vi) Discuss the significance of inplace parameter in Series objects? 1
iv) What is Broadcasting? 1
v) Discuss how to add labels, titles, and legends to charts in Matplotlib. 1

SECTION-B

Q2. WAP to create a figure containing two subplots where one subplot displays a line chart and the other displays a scatter chart.

Q3. Discuss ways in which missing, duplicate and categorical values can be handled.

Q4. Write a program to read a CSV file into a DataFrame and print the following attributes of the DataFrame object: index, columns, shape, and size.